

Pharmacokinetic and Pharmacodynamic Analysis of the Tirzepatide 2-Compartment Model

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Abstract

Tirzepatide, a groundbreaking therapeutic agent in the management of type 2 diabetes and obesity, functions through a novel dual agonist mechanism. This dual action not only enhances glycemic control but also significantly contributes to weight loss, leveraging mechanisms that regulate appetite and insulin secretion. Given the complexity of its pharmacokinetics (PK) and the broad implications of its pharmacodynamics (PD), understanding tirzepatide's behavior within the body is crucial for optimizing therapeutic strategies. A two-compartmental model class, written in matlab, was implemented to simulate the pharmacokinetics and pharmacodynamics of tirzepatide for variable parameters and dosing regimes.

Keywords: Pharmacokinetics, pharmacodynamics, systems pharmacology, tirzepatide, weight-loss

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1. Introduction

Obesity is an ongoing challenge that proves accessible, feasible, and effective methods of controlling and counteracting its progression are necessary. Obesity is defined as a chronic medical condition where a person's weight is higher than what can be considered normal, or healthy, for a given height [7]. Obesity is a global epidemic affecting both developed and developing countries. Worldwide obesity has nearly tripled since 1975. In 2016, more than 1.9 billion adults, 18 years and older, were overweight, and of these over 650 million

were obese [10]. The obesity epidemic is quite extensive, especially in the United States where 41.9% of adults in 2023 were considered as having obesity [6]. Rates are also climbing for younger generations, where nearly 20% of children and adolescents aged 2-19 years old are obese [6]. As a leading cause of death, obesity contributes to at least 2.8 million deaths around the world each year.

Although the causes for obesity are multifaceted and tough to properly identify, several factors such as environmental, hereditary, and lifestyle variables have prominent implications in the development of obesity [5]. Caloric intake that exceeds caloric expenditure is a fundamental factor, but the condition is also strongly influenced by the environment and genetic predisposition [3]. Factors such as a diet high in fats and sugars, and a sedentary lifestyle, contribute significantly to its prevalence [3]. The pathophysiology of obesity involves multiple systems and is not merely a result of overeating [3]. Changes in the endocrine function, particularly in insulin and leptin signaling, play crucial roles in fat accumulation and metabolic alterations in obese individuals [9]. A number of complications may also result from obesity. Chronic illnesses such as diabetes mellitus, hypertension, cardiovascular disease, and stroke are typical complications [5]. It also contributes to musculoskeletal disorders like osteoarthritis [9]. Tackling the overweight issue through simple lifestyle changes proves to be very difficult since they are not typically satisfactory. Current treatment strategies for obesity range from lifestyle modifications, such as increased physical activity and dietary changes, to pharmacotherapy and surgery [11]. Medications approved for long-term use include orlistat, lorcaserin, and phentermine-topiramate, among others [11]. Bariatric surgery may be considered for severe obesity when other interventions have failed [11]. Therefore, surgery and medicine are often attractive ways to lose weight.

A typical screening tool for obesity is the Body Mass Index (BMI). A BMI of 27-30 kg/m² classifies as overweight and a BMI over 30 kg/m² classifies as obesity [8]. Obesity is then separated into three classes: Class 1 obesity is a BMI of 30-35 kg/m², class 2 is a BMI of 35-40 kg/m², and class 3 (severe) as a BMI over 40 kg/m² [7]. However, because BMI takes only weight into consideration, it does not acknowledge increased adiposity or the distribution of body fat. Thus, a six-foot man with a BMI of 32 indicates obesity, but perhaps this man is a weight lifter and has excessive muscle rather than body fat, making him overall healthy. As a result, using BMI as an indication of health is no longer used as the only method.

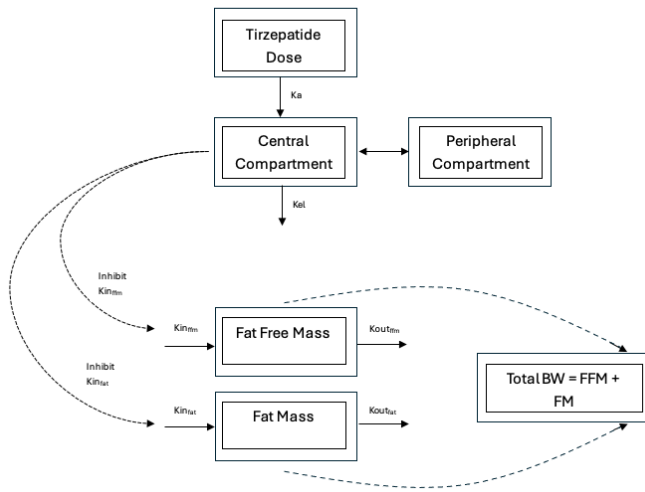


Figure 1. Schematic of the Pharmacodynamic Model of Tirzepatide

Obesity treatment is challenged by high rates of non-compliance and the limited effectiveness of current drugs. Behavioral modifications are difficult to maintain, and many patients regain weight after discontinuing treatment. Access to surgery and advanced pharmacotherapies is also limited by economic and healthcare system barriers. Tirzepatide represents a novel approach in the pharmacological treatment of obesity. Tirzepatide is a dual agonist of the glucagon-like peptide-1 (GLP-1) and glucose-dependent insulinotropic polypeptide (GIP) receptors [5]. These incretin hormones play vital roles in glucose homeostasis and appetite regulation [1]. GLP-1 receptor agonists are already well-established in the treatment of type 2 diabetes (T2DM) and obesity due to their ability to enhance insulin secretion, inhibit glucagon release, and promote satiety [1]. This hormone is especially useful as it lowers blood glucose levels in a glucose-dependent manner. When glucose levels in the blood are high, GLP-1 enhances the secretion of insulin, pushing the glucose levels down. Additionally, GLP-1 slows gastric emptying, reducing the sensation of hunger [1]. The addition of GIP receptor agonism enhances these effects, potentially leading to greater efficacy in weight reduction and glucose control [1]. GIP also plays a role in promoting energy storage and in lipid metabolism. By stimulating both of these hormones, tirzepatide improves glycemic control and promotes significant weight loss.

2. Methods

2.1. Pharmacokinetic Model

Equations 1 describe the pharmacokinetics of tirzepatide in the central and peripheral compartment. Equation 1a describes the central compartment concentration; equation 1b describes the peripheral central compartment concentration; equation 1c describes the concentration leaving the virtual dose compartment; and equation 1d describes the concentration in the virtual clearance department.

$$\frac{dy_C}{dt} = \frac{k_a F}{V_C} y_D - k_{el} y_C - k_{CP} y_C + \frac{k_{PC} V_P}{V_C} y_P \quad (1a)$$

$$\frac{dy_P}{dt} = \frac{k_{12} V_C}{V_P} y_C - k_{21} y_P \quad (1b)$$

$$\frac{dy_D}{dt} = -k_a y_D \quad (1c)$$

$$\frac{dy_{CL}}{dt} = (1 - F) k_a y_D + k_{CL} V_C y_C \quad (1d)$$

The equations in 1 are based off the schematic provided in pg. 65 of [4].

Rate Constants in the Pharmacokinetic Model

k_a : Absorbance (1/hr)

k_{CL} (CL/ V_C) : Clearance (1/hr)

k_{CP} (Q/ V_C) : Central to Peripheral (1/hr)

k_{PC} (Q/ V_P) : Peripheral to Central (1/hr)

Model parameter estimates provided in table 1.

2.2. Pharmacodynamic Model

In addition to the equations provided in 1, the PD model also adds weight mechanic equations for fat-free mass (FFM) and fat mass (FM) as listed in Equations 3. Both compartments are modeled in the schematic shown in page 74 of [4]. The central compartment concentrations inhibit the K_{in} of both the FFM and FM compartments. This inhibition is modeled by the following linear equation:

$$k_{in,t} = k_{in,baseline} - k_{in,baseline}(placebo + slope \times y_{C,t}) \quad (2)$$

Where t represents the value at that given time step. The baseline, placebo, and slope values are provided in table 2.

$$\frac{dy_{FFM}}{dt} = k_{in,FFM} - k_{out,FFM} y_{FFM} \quad (3a)$$

$$\frac{dy_{FM}}{dt} = k_{in,FM} - k_{out,FM} y_{FM} \quad (3b)$$

Key Variables of the PKPD model

k_a : Absorbance (1/hr)

CL : Clearance (L/hr)

V_C : Central Volume of Distribution (L)

V_P : Peripheral Volume of Distribution (L)

BMI : Body Mass Index (kg/m²)

Key Metrics of Pharmacodynamic Model

AUC_{central} : Area under the curve of the central compartment concentration
% Change : The expected % change in bodyweight, FFM and/or FM

2.3. Generating Virtual Populations

In order to implement the pharmacodynamic model, virtual populations are required to get BMI, FFM, and FM values.

BMI for patients is randomly generated using the following algorithm and the values provided in table 3. To generate BMI values, simply use the following equation

$$BMI = Z\sigma_{BMI} + \mu_{BMI}$$

for each individual, where Z is the standard normal distribution and σ_{BMI} and μ_{BMI} are the standard deviation and mean of the population BMI. If any individual has a BMI below the cutoff mentioned in 3, re-run this equation until all individuals are above the cutoff. Once a virtual sample is generated, weights can be generated for the population using equation 4 where Z is the standard normal distribution $\mathcal{N}(0, 1)$ and p is a correlation parameter ranging from 0 to 1. For male and female cohorts¹, the values of $p = 0.76$ and $p = 0.85$ are used, respectively [2]. Z_{BMI} refers to the standardized distribution of the BMI values collected. Once standardized body weight values are assigned, it's important to then re-scale values by body weight mean and standard deviation.

¹correlation values were collected from a sample of Nigerian patients

$$BW = pZ_{BMI} + \sqrt{1 - p^2}Z \quad (4)$$

To then get the fat-free and fat masses of each patient, the allometric equations 5 obtained from 76 of [4] can be used.

$$FFM = \begin{cases} \frac{9270 \cdot BW}{6680 + 216 \cdot BMI} & \text{if sex = male} \\ \frac{9270 \cdot BW}{8780 + 244 \cdot BMI} & \text{else female} \end{cases} \quad (5)$$

The mean, standard deviation and cutoff for BMI and body weight are provided in 3.

2.3.1. Individual Parameters

As given by [4] in page 58, we can then use the equations 6a and 6b to get individual clearance and volume of distribution values.

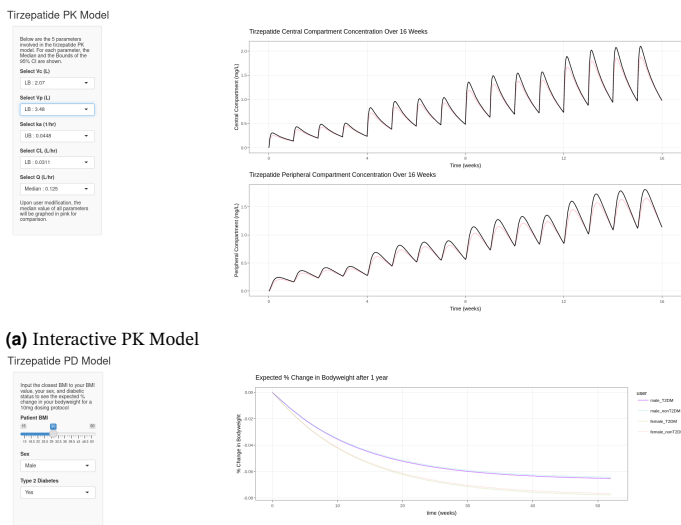
$$iCL = popCL \left(\frac{BW}{70} \right)^{0.8} \quad (6a)$$

$$iV_d = popV_d \frac{FFM + 0.487FM}{70} \quad (6b)$$

where $popCL$ and $popV_d$ are the population estimates of clearance and the respective volumes of distribution, found in table 1.

2.4. Interactive Shiny Apps

For the interactive visualization of our tirzepatide PK model, we chose to display the concentration of tirzepatide in the central and peripheral compartments as well as the cumulative dosing over 16 weeks. Through the user interface we chose to modify five parameters: V_c (L), V_p (L), k_a (1/hr), CL (L/hr), and Q (L/hr). Values for these parameters can be modified between the lower bound value, median, and upper bound value. The default plot shows the median of all parameters in black, while the plots with altered parameters appear in pink. Similarly, for the interactive visualization of our tirzepatide PD model, we chose to display the expected percent change in body weight after a year of monotonic dosing, for four different groups: male and female T2DM and normal groups. Through the interface, the user can select a specific BMI value, which sex, and whether or not they have T2DM. The app's display shows what the percent change in body weight looks like for different BMI values.



(b) Interactive PD Model

Figure 2. Interactive Models Screenshots of the shiny apps are shown for the PK & PD model shown in panel a and b, respectively. The links to these sites can be accessed from the resources section 7

3. Pharmacokinetic Results

3.1. Single Dosage

The single-dose simulations in figure 3 show the curves of each individual injection. With higher dose injection, the absorption and clearance processes in the central compartment and peripheral compartment are both significantly accelerated. Correspondingly, the process of drugs leaving the subcutaneous region and being absorbed is also vastly accelerated. Panel (c) of figure 3 shows that tirzepatide fully distributes into the body after 1 week, supporting the weekly dose recommendation for tirzepatide. There are notable artifacts in the mass balance, albeit small, but should be acknowledged. Currently the implementation defaults to ode45, but conversion to ode78 prevents this at the cost of expensive operations.

`1 Model.simulatePK(p, y0, dose, "solver", @ode78)`

Code 1. generateBMI Function. Detailed context can be found in the Model.m file located in the provided GitHub. (See *RESOURCES*)

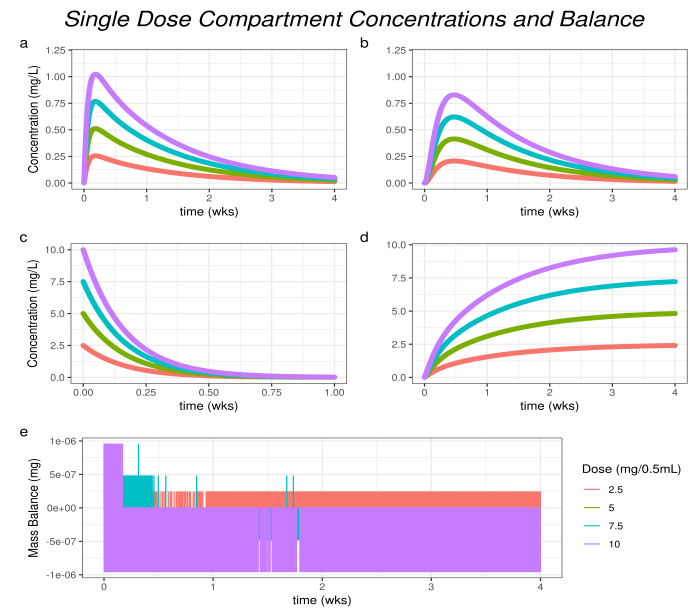


Figure 3. Single Dosage Effect of single dosing on all 4 compartments described in equations 1 for 2.5, 5, 7.5, and 10 mg/0.5 mL dosing over 4 weeks. The concentrations in the (a) central, (b) peripheral, (c) dosage, and (d) clearance compartments are shown. Panel (e) shows the mass balance for each dose regime. Note that panel (c) has been rescaled to 1 week.

3.2. Steady State

To identify time when steady state was reached, a 20 week monotonic dosing schedule was plotted for 2.5-15 mg doses, where the first peak to have a value close to the final peak concentration at week 20 was labelled as the week at which steady state was achieved.

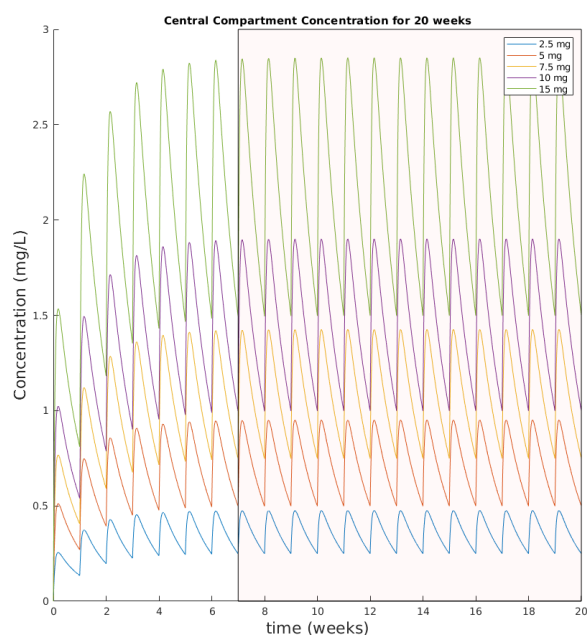


Figure 4. Monotonic Steady State Dosing Monotonic weekly dosing from 2.5-15 mg in 2.5 mg increments are shown for the central compartment over a 20 week duration. Steady state is achieved after 7 weeks of dosing.

3.3. Missed Dose Analysis

Tirzepatide is injected subcutaneously once weekly to have a significant performance. Figure 4 shows that the peaks of every dosing regime line up exactly, so we can focus on the relative concentration to get an idea of the behavior for any dosing regime. In this case, we simulated up to steady state and then did a skip dose analysis. In the unforeseen case of missing a dose, we simulated what the relative concentration of Tirzepatide in the body looks like to compare it to consistent dosing in Figure 5. A skipped dose is designated as being seven days late, and this evidently shifts the tirzepatide concentration down until the next dose is administered. Eventually with consistent dosing, the trend continues on the upward trend and stabilizes within the red region, which is indicative of how standard dosing looks like. This simulation proves that if a tirzepatide dose is missed, it should be administered as soon as possible within 4 days after the day of missed dose. Administration of tirzepatide dose within 4 days of the missed dose resulted in $< 20\%$ increase in $C_{\max}$ associated with the subsequent scheduled dose. If more than 3 days have passed, it is recommended that the missed dose be skipped and to resume the regular weekly (QW) dosing schedule by administering tirzepatide on the next regularly scheduled day.

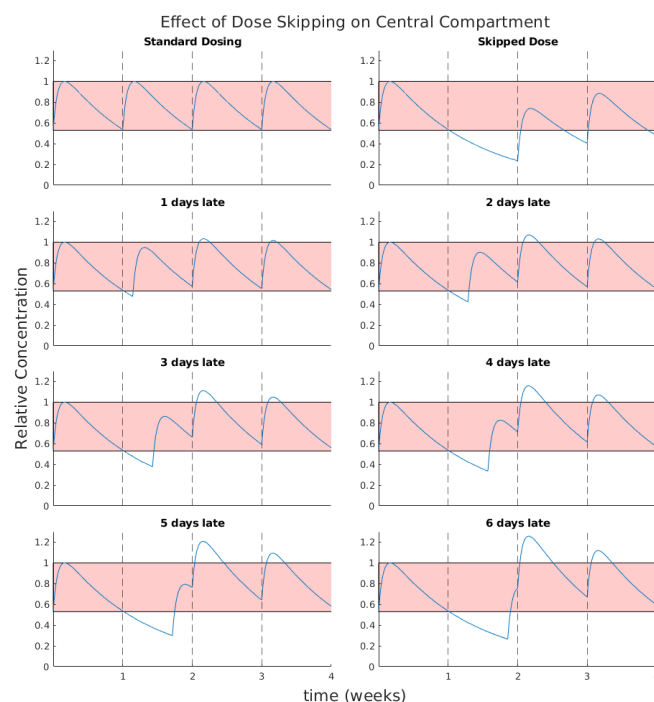


Figure 5. Missed Dose Analysis The effect of dose skipping for each possible day of the week is shown for relative concentrations in the central compartment. Dosing is recommended within 3 days or else wait until the next week. Relative concentration is taken by dividing concentration by the maximum steady state concentration.

3.4. AUC of Central and Peripheral Compartment

From [4], the current dosing regime is to up-titrate 2.5 mg doses every 4 weeks until the patient reaches 10 mg in week 12 unless adverse side effects are observed. Figure 6 shows the AUC distributions in the central and peripheral compartments over a 16 week period with the respective up-titration every 4 weeks (eg., 2.5, 5, 7.5, and 10 mg). All 243 combinations of PK values from Table 1 were used to generate this distribution. The distribution of Figure 6 is directly provided below.

V_d	Min	LQ	Median	UQ	Max
Central	1.24	1.301	1.344	1.392	1.428
Peripheral	1.241	1.28	1.323	1.367	1.412

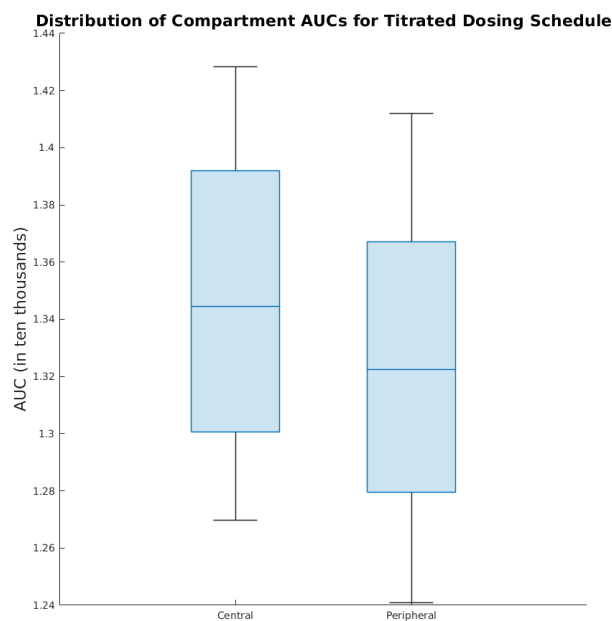


Figure 6. Central & Peripheral Compartment AUCs The AUC distributions of the central and peripheral compartments for a titrated dosing schedule as described in section 3.4. The reference AUC is calculated by taking the individual AUC and dividing it by the AUC of the median values provided in table 1

3.5. Global Sensitivity Analysis

Because the only individual patient parameters are clearance and volume of distribution (VOD), global sensitivities were performed on clearance and each respective (VOD), as well as the volumes of distribution against each other. Global sensitivity analysis helps in assessing the robustness of a model by identifying parameters that significantly influence the model's predictions. This understanding is important for validating the model against experimental or clinical data. The heat map in figure 7 uses the colors blue (lower sensitivity) and red (higher sensitivity) to display the central compartment AUC. Lower sensitivity values indicate the parameter in question has less of an impact on the model. The opposite is true of the higher sensitivity values.

What we conclude is that a lower volume of distribution and a lower clearance value increases sensitivity. This conclusion is reasonable as we can justify the decreased volume of distribution leading to an increased concentration of tirzepatide within the respective compartment and decreasing clearance increases the duration of time that the drug stays within the compartment.

4. Patient Distribution Analysis

Using the methods described in 2.3, 1000 virtual patients were generated for both the type 2 diabetic (T2DM) patient and non-diabetic (normal) patient groups. We identified the central and peripheral volumes of distribution and clearance values (as mentioned in section 2.3.1) as the primary factors that vary within the population. From section 2.3, we used a correlation parameter ρ from [2], which was collected from a Nigerian population.

4.1. BMI Distribution by Status and Sex

Distributions of BMI by sex and diabetic status are shown in figure 8, with summary statistics from the box plots located in table 4. The only significant difference in populations stems from diabetic status rather than sex.

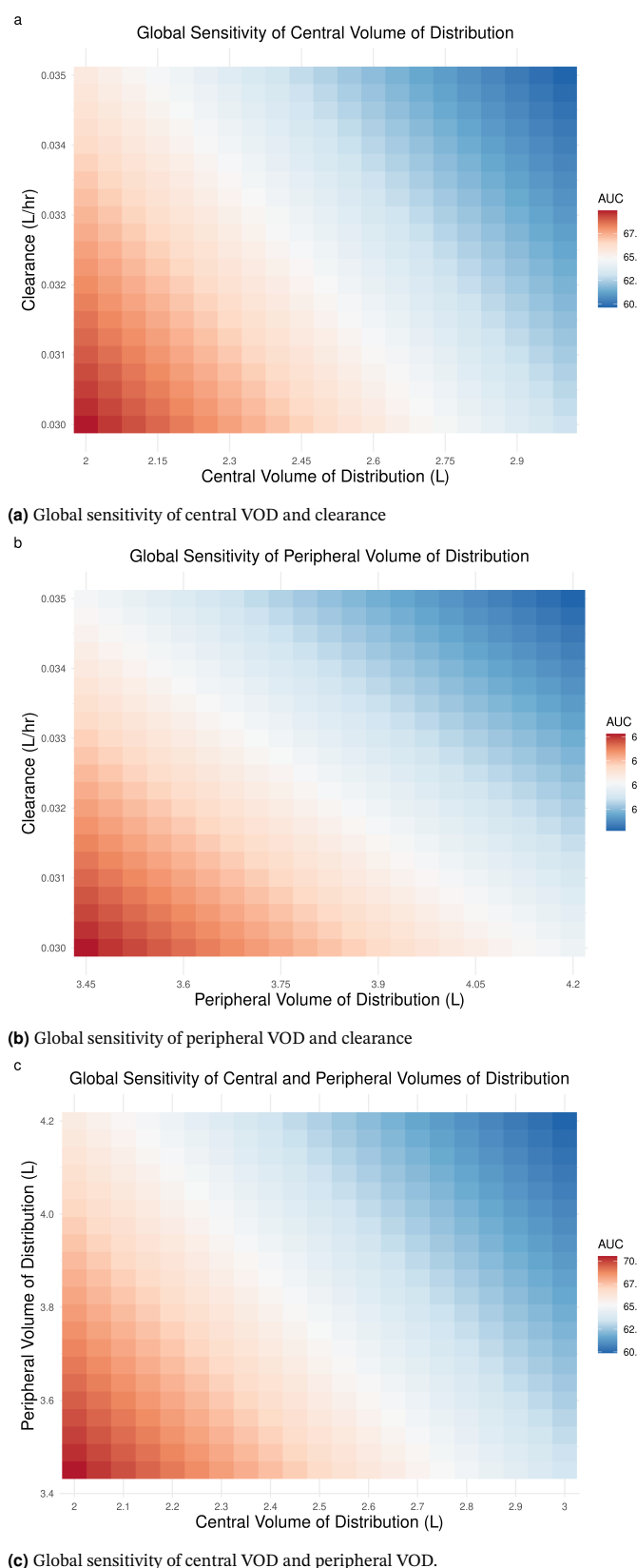


Figure 7. Pharmacokinetic Global Sensitivity Analysis Global sensitivity analysis of the central VOD, peripheral VOD, and clearances are shown with respect to each other. The AUC of the central compartment concentration over one year was used as the metric for this sensitivity analysis.

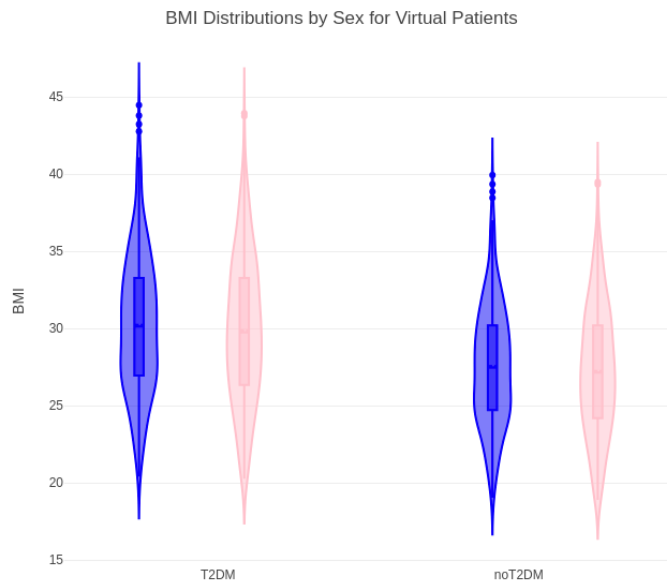


Figure 8. BMI Distribution BMI distributions by sex and diabetic status for 1000 virtual patients in both the T2DM and non-T2DM group. Summary statistics can be found in table 4

4.2. Allometric Relations Between Parameters and BMI

For future use and ease of simplicity, linear approximations were desired between individual parameters (found using the methods discussed in 2.3.1) and BMI. figure 9 provides a visualization showing the spread of individual parameters against BMI with their linear fits. The lines of best fit are provided in table 5.

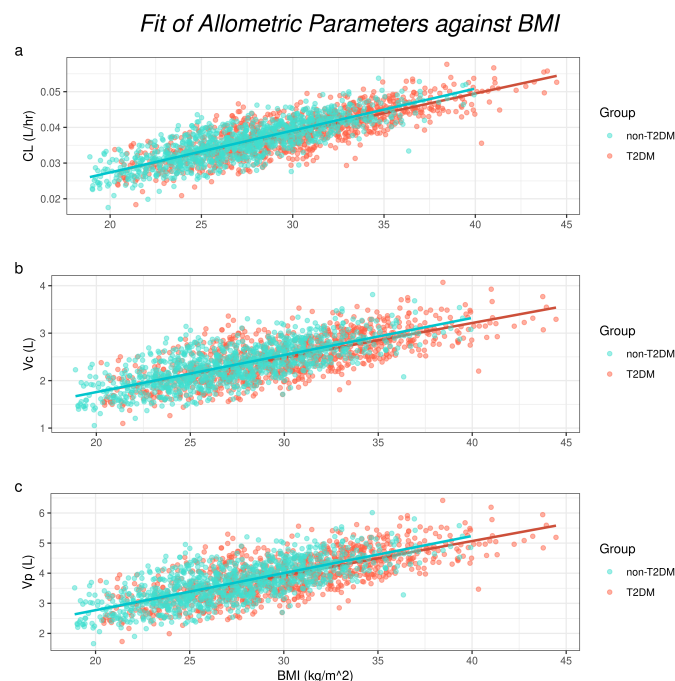


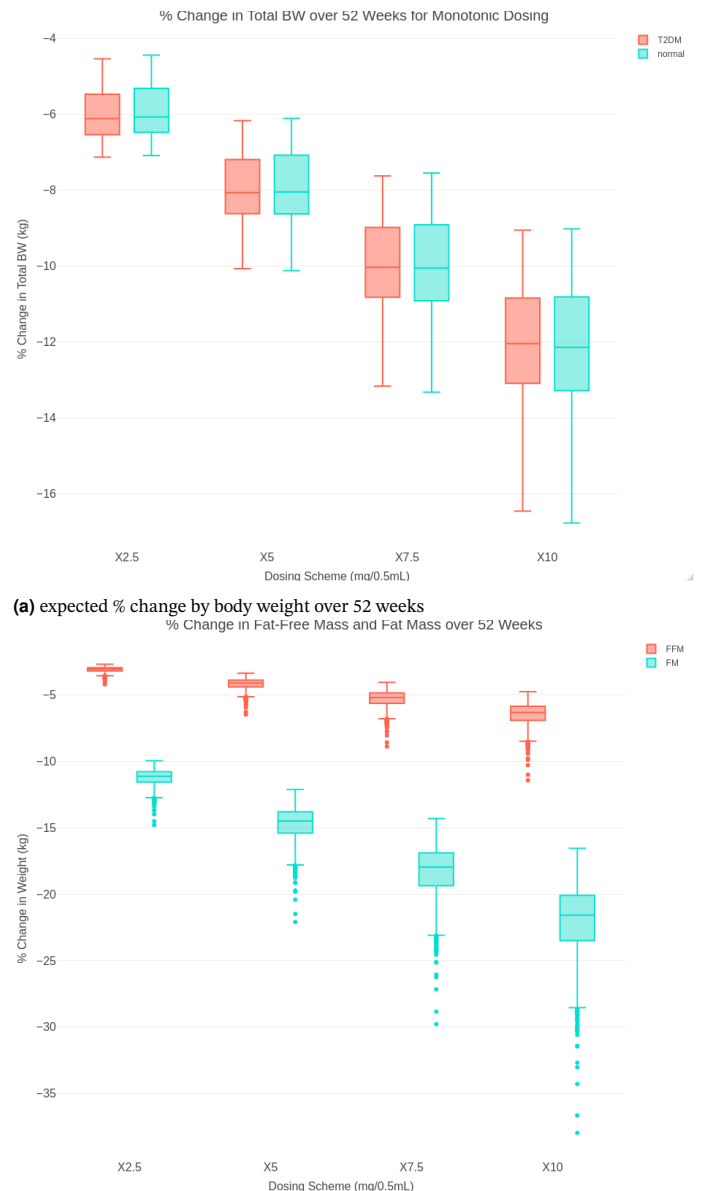
Figure 9. Inter-Individual Variability of Allometric Parameters Individual clearances (CL) and volumes of distribution (V_C and V_P) are shown for both the non-diabetics and diabetic groups with respect to BMI.

Additional details about the effect of varying any two of the allometric key parameters are discussed in the sensitivity analysis of section 5.

5. Pharmacodynamic Results

5.1. Expected Change in Body Weight

Below is the expected percent change in body weight after 1 year of monotonic dosing for 2.5, 5, 7.5, and 10 mg protocols. Each cohort has 1000 individuals and the values are shown here. This plot lines up with current literature that suggests about a 6% drop in weight with 2.5 mg with a nearly linear drop down to 12% drop in weight at 10 mg. Interestingly, we also see that the percentage loss in fat mass over the year is far greater than the percentage loss in fat-free mass. Going by median values, there is an average of 3.5x difference in the percentage loss in fat mass over fat-free mass. Additionally the spread of fat-free mass loss is quite low, compared to the fat mass loss, which even at the lowest values, is still more than 1.5 times the difference of the maximum values in the fat-free mass distributions.



(b) expected % change by fat-free (FFM) and fat (FM) mass over 52 weeks

Figure 10. Population Weight Change The percent weight drop from baseline for monotonic dosing over 52 weeks is shown. Panel a shows the total bodyweight lost over the year across the distribution of both T2DM and non-T2DM virtual patients. Panel b displays the same information, but now splits the fat-free and fat mass, showing the difference in weight loss based on mass type.

5.2. Global Sensitivity Analysis

A global sensitivity analysis (shown in figure 11) was additionally performed on the pharmacodynamic outputs, focusing on the percentage change from a baseline bodyweight of 100 kgs after one year of tirzepatide usage. All other parameters were set to the median reference estimate.

We can see from figures 11a and 11b that VOD has no effect on the expected weight change from baseline, but clearance does. This is useful information as it allows us to make general doses for the public without worrying about how individual VODs will affect dosing. Figure 11c was plotted for consistency, but from the colorbar, it should be clear that there is a miserly effect on actual weight loss, which matches with the conclusions we made with figures 11a and 11b.

6. Conclusion

The pharmacokinetic simulation for a single injection shows that with increasing concentration, there is a significant increase in drug absorption and clearance rate in both central and peripheral compartments. However, the drug concentration curves for both central and peripheral compartments corresponding to this single injection are evidence tirzepatide is more effective with multiple injections.

The pharmacokinetic simulation for a weekly injection shows a trend of decreasing magnitude of each local concentration peak. Each new injection enters a steady state after 7 weeks. Such saturation tendency provides strong proof for increasing the weekly dose every 4 injections for patients.

The pharmacokinetic simulation for missed dose shows that taking the weekly injection less than 4 days late provides less than 20% increase in C_{max} . Meanwhile, a more than 20% increase in C_{max} is evident if the weekly dose is more than 3 days late. There is strong evidence that taking the weekly injection more than 3 days late is not recommended. It is more effective to just skip the injection and resume regular weekly injection the following week. Otherwise, an unnecessarily high C_{max} can post unbearable harm to patients.

The global sensitivity analysis suggests that the AUC of the current clinical dosing regime shows lower sensitivity for higher VOD (both central and peripheral) and a higher clearance coefficient. Both of these metrics are normally related to a high BMI, suggesting that the effect of tirzepatide is more stable for obesity patients than normal users. Thus, this data can be used as proof in marketing tirzepatide as a weight loss medicine.

The pharmacodynamic simulation for expected weight loss shows a 6% decrease in weight for the 2.5 mg/week injections and a 12% decrease in weight for the 10 mg/week injections. The simulation also shows the percentage decrease in fat mass has a higher magnitude than that in fat-free mass, which verifies that the weight loss is mostly due to the loss of fat.

The expected weight loss global sensitivity analysis suggests that the expected weight loss after using tirzepatide for a year has a lower sensitivity for higher clearance coefficient. Additionally, there is almost no change for variation in central and peripheral VOD. However, the global heat map also shows a complex sensitivity if both VOD are changing. The map demonstrates some tendency of lower sensitivity with bigger VOD, but it can be generated due to statistical uncertainty in the cited literature.

7. Links & Resources

Links:

 GitHub

 PK Interactive

 PD Interactive

 vboosha1@jhu.edu, ykluczn1@jhu.edu, ygao129@jhu.edu

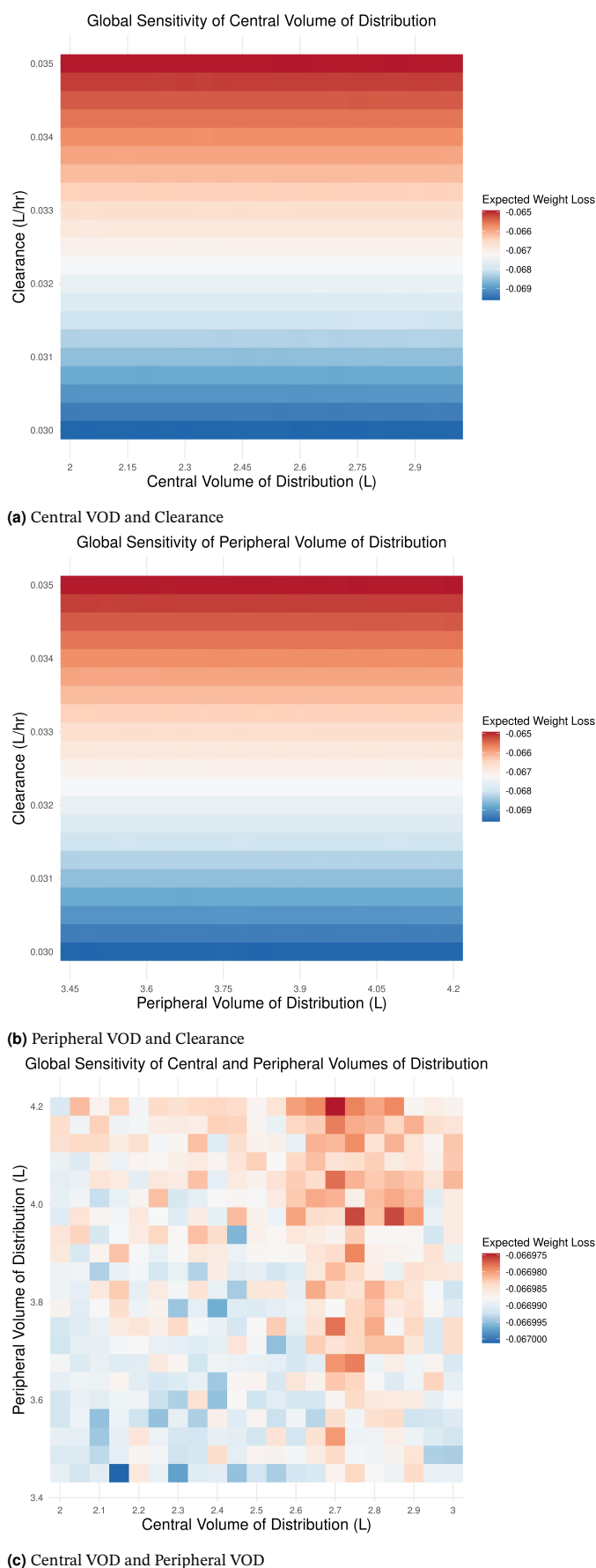


Figure 11. Pharmacodynamic Global Sensitivity Analysis Global sensitivity of the central VOD, peripheral VOD, and clearances are shown with respect to each other. The expected weight lost over one year was used as the measure of sensitivity, a key metric described in section 2.2

Reflections

In the simulation of tirzepatide's PK-PD model, our data was primarily focused on adults with type 2 diabetes mellitus. This population is typical in clinical trials aimed at assessing the efficacy of a drug awaiting regulatory approval. To stimulate a broader patient demographic, we generated a virtual patient pool with varying BMI and prevalence of T2DM cohorts. In evaluating a variety of patients with or without T2DM, as well as with different BMI, we were able to evaluate the drug's impact across demographics. Given the target market for this drug involves a spectrum of body weights and compositions, these simulations were vital in assessing the drug's pharmacological behavior.

Certain populations were excluded from our study, including pediatric and geriatric individuals. There groups often require separate studies due to distinct physiological profiles that can impact a drug's PK-PD qualities. For instance, these groups often have major differences in body composition and metabolic processes which can result in variations in how a drug is absorbed, distributed, metabolized, and cleared. Moreover, tirzepatide's safety and efficacy may vary among these excluded populations. The elderly may have an increased risk of hypoglycemia while pediatric patients are still undergoing development and have other metabolic needs.

In reflecting on the populations and data used in our modeling process, the importance of considering a wide variety of physiological variables is highlighted. The applicability of PK-PD models is enhanced with a broad array of demographics tested. Future studies should aim to include the aforementioned excluded populations to refine the accuracy of the model and provide increased data on tirzepatide's use across more potential users.

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8. Appendix

8.1. Pharmacokinetic Parameters

Table 1. Pharmacokinetic Parameter Estimates

Parameter	Median and 95% CI	Summary
F	0.8 (fixed)	Bioavailability
k_a	0.0363 (0.0287, 0.0448)	Absorbance Rate (1/hr)
CL	0.0326 (0.0311, 0.0341)	Clearance (L/hr)
Q	0.126 (0.101, 0.143)	Inter-compartmental Clearance (L/hr)
V_C	2.48 (2.07, 2.98)	Central Volume of Distribution (L)
V_P	3.91 (3.48, 4.17)	Peripheral Volume of Distribution (L)

Note: Obtained from pg. 58, 2023. [Online].
Available: https://www.accessdata.fda.gov/drugsatfda_docs/nda/2022/215866Orig1s000ClinPharmR.pdf

8.2. Pharmacodynamic Parameters

Table 2. Pharmacodynamic Parameter Estimates

Parameter	Estimate and 95% CI
Baseline fat-free mass (kg)	65.2 (64.7, 65.7)
Baseline fat mass (kg)	30.5 (29.9, 31.3)
First-order elimination rate constant, K_{out} (1/wk)	(0.0797, 0.0743, 0.0851)
Slope for drug effect reducing fat-free mass K_{in} (kg/wk)	3.71(3.62, 3.92)(10^{-5})
Slope for drug effect reducing fat mass K_{in} (kg/wk)	1.19(1.16, 1.26)(10^{-4})
Placebo fractional reduction in fat-free mass K_{in}	0.0205 (0.0191, 0.0219)
Placebo fractional reduction in fat mass K_{in}	0.0796 (0.0741, 0.0847)

Note: Obtained from pg. 76, 2023. [Online].
Available: https://www.accessdata.fda.gov/drugsatfda_docs/nda/2022/215866Orig1s000ClinPharmR.pdf

8.3. Patient BMI & Weight Distributions

Table 3. Mean and Standard Deviation of BMI and body weight given Type 2 Diabetes Mellitus

Parameter	T2DM	mean	standard deviation	cutoff
BMI	YES	29.7	4.98	20.2
	NO	27.1	3.44	18.8
BW	YES	86.2	17.2	56.6
	NO	79.5	15.6	50.7

Note: Obtained from pg. 55, 2023. [Online].
Available: https://www.accessdata.fda.gov/drugsatfda_docs/nda/2022/215866Orig1s000ClinPharmR.pdf

8.4. BMI Summary Statistics

Table 4. Summary statistics for BMI distributions in figure 8

T2DM	sex	Mean	Median	StDev	Min	Max
YES	M	30.21	30.12	4.46	20.3	44.46
YES	F	29.86	29.74	4.74	20.26	43.96
NO	M	27.55	27.46	3.89	19.02	39.93
NO	F	27.24	27.13	4.12	18.89	39.49

Table 5. Lines of best fit for the diabetic and non-diabetic groups for panels a, b, and c, respectively, in figure 9

Parameter	T2DM	non-T2DM
CL (L/hr)	$0.001107 \cdot BMI + 0.005207$	$0.001174 \cdot BMI + 0.003904$
V_c (L)	$0.072830 \cdot BMI + 0.301250$	$0.078140 \cdot BMI + 0.195320$
V_p (L)	$0.114800 \cdot BMI + 0.474900$	$0.123200 \cdot BMI + 0.307900$